BUILD VS. BUY: SCALING FRAUD DETECTION IN DIGITAL LENDING

A product decision framework | Independent research and proposal

THE DECISION

As a digital lending product scales, fraud vectors evolve faster than manual review can keep pace with. At some point, every lending product team faces the same strategic fork: build an in-house fraud detection capability, or buy/integrate a third-party solution. This is a framework for making that call — and for what happens after it's made.

WHY THIS ISN'T JUST A TECHNICAL QUESTION

Build vs. buy gets treated as an engineering decision when it's really a product and business one. The wrong call in either direction is expensive: buying locks you into a vendor's roadmap and blind spots; building burns 2-3 quarters of engineering capacity on a problem that proven vendors have already solved, delaying the actual product differentiation work.

EVALUATION FRAMEWORK

- **Time to market:** how fast fraud losses need to be contained versus how long a build would realistically take.
- **Cost of ownership:** vendor licensing and integration cost versus fully-loaded build cost (engineering time, ongoing maintenance, model retraining).
- **Compliance & data control:** data residency and audit-trail requirements that a vendor must demonstrably meet, not just claim.
- **Strategic differentiation value:** whether fraud detection is a genuine competitive differentiator for this product, or table-stakes infrastructure.
- **Engineering opportunity cost:** what a 2-3 quarter build actually displaces on the roadmap, and whether that trade-off is worth it.
- **Vendor lock-in exposure:** how hard it becomes to leave a vendor once transaction history and models are embedded in their system.

THE RECOMMENDATION: A HYBRID PATH

Buy a proven rules-based engine for baseline fraud detection — this is largely solved, well-audited technology, and building it from scratch mainly delays time-to-market without adding differentiation. In parallel, build a thin proprietary layer on top using first-party behavioral and transaction data unique to the product — this is where genuine differentiation lives, and it's a far smaller build than a full detection engine.

This isn't a permanent split. It's a staged path: launch fast and compliant with the vendor baseline, redirect the engineering capacity saved from not building that baseline into the proprietary layer, and revisit full in-house ownership only once transaction volume and data maturity justify the investment.

SELLING THE RECOMMENDATION ACROSS STAKEHOLDERS

- **Engineering:** frees 2-3 quarters of capacity for the proprietary layer instead of rebuilding commodity infrastructure.
- **Compliance/Risk:** a vendor with an existing audit trail and compliance certification reduces regulatory risk versus an unproven in-house model.
- **Finance:** licensing cost is predictable and front-loaded; a full build carries multi-quarter opportunity cost that's harder to quantify but very real.
- **Leadership:** ships fraud protection in weeks instead of quarters, without giving up a long-term differentiation path.

12-18 MONTH VISION

Phase 1 (0-3 months): Launch on the vendor baseline, instrument every decision for data collection. Phase 2 (3-9 months): Ship the proprietary behavioral layer as a second signal alongside the vendor's, measuring lift. Phase 3 (9-18 months): If the proprietary layer consistently outperforms the vendor baseline on catch rate, begin a staged transition to reduce vendor dependency — informed by real data, not a decision made upfront on assumptions.

SUCCESS METRICS

- **Fraud catch rate:** of confirmed fraud identified before loss, tracked separately for vendor baseline vs. proprietary layer.
- **False-positive rate:** legitimate customers incorrectly flagged — the real cost of an overly aggressive model.
- **Time-to-decision:** time from transaction to fraud/no-fraud decision, since latency directly affects customer experience.

- **Cost per transaction:** total cost per transaction reviewed, vendor fees plus engineering cost, tracked over time to validate the phased plan.